

09 — Risk Register

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

Field	Value
Document ID	RR-001
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Related docs	Charter , Compute Memo , Roadmap

Risk Assessment Framework

Each risk is assessed on:

Dimension	Scale
Likelihood	Low / Medium / High
Impact	Low / Medium / High / Critical
Overall severity	Low / Medium / High / Critical

1. Scientific Risks

ID	Risk	Likelihood	Impact	Severity	Mitigation	Fallback
SR-01	k-mer features are insufficient to capture regulatory signal; models learn GC content or simple composition instead of biologically meaningful patterns.	Medium	High	High	Feature ablation experiment (remove GC, test remaining signal). Compare k-mer importances to random expectations. Validate top features against JASPAR.	Add motif scan features. Move to CNN (Phase 2) which learns features directly.

SR-02	ENCODE cCRE labels are noisy or circular: cCREs are themselves computationally predicted, so the model may learn to replicate the ENCODE pipeline rather than discover new patterns.	Medium	Medium	Medium	Use multiple label sources (DNase peaks, histone marks) as independent validation. Clearly state limitation in publications.	Frame results as "consistent with ENCODE annotations" rather than "discovered new biology."
SR-03	Negative set design introduces bias: if negatives are systematically different from positives (e.g., in sequence complexity, repeat content), the model exploits these artifacts.	Medium	High	High	GC-matched + exclude-known negatives. Multiple negative sampling strategies with sensitivity analysis (E7). Check feature importance for compositional vs. motif features.	If all negative strategies show same features, the bias is minimal. If results vary wildly, report the sensitivity analysis honestly.
SR-04	Motif databases are incomplete: top features may not match known motifs because the motifs are novel, not because they are noise.	High	Low	Medium	Report unmatched motifs as "candidate novel patterns" with appropriate caveats. Do not claim biological significance without experimental validation.	Treat this as a positive research finding if the patterns are consistent and reproducible.
SR-05	Cell-type-specific results may not generalize: models trained on K562 may not transfer to other cell types, limiting biological insight.	Medium	Medium	Medium	Cross-cell-type experiments (E4). Report per-cell-type and pooled results. Document which findings are cell-type-specific vs. general.	If cell-type-specific, this is actually an interesting result about regulatory diversity.

SR-06	Overfitting to training chromosomes: despite chromosome-holdout, the model may learn chromosome-specific patterns that don't generalize.	Low	Medium	Low	Verify that test chromosomes (chr19–22) have representative regulatory element density. Add leave-one-chromosome-out analysis as sanity check.	If test-set performance is much lower than CV, investigate chromosome-specific confounders.
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2. Data Risks

ID	Risk	Likelihood	Impact	Severity	Mitigation	Fallback
DR-01	ENCODE portal becomes unavailable or changes data access policy.	Low	High	Medium	Download and cache all required files early. Store versioned copies as Kaggle datasets. Document exact file URLs and checksums.	Use cached data. If ENCODE is permanently unavailable, switch to Roadmap Epigenomics or FANTOM5 (alternative sources).
DR-02	Reference genome version mismatch: using hg38 annotations with wrong genome build causes incorrect sequence extraction.	Low	Critical	Medium	Always verify genome assembly version. Include assembly version in data pipeline logging. Cross-check extracted sequences against UCSC Genome Browser for sample regions.	Detect early by checking sequence content at known loci.

DR-03	Data too large for Kaggle: curated dataset exceeds Kaggle storage limits.	Low	Medium	Low	Start with K562-only subset (~100K regions). Upload per-chromosome FASTA as separate datasets. Use compressed formats (.npz).	Process locally and upload only feature matrices to Kaggle.
DR-04	Class imbalance: far more regulatory or non-regulatory regions depending on sampling strategy.	Medium	Medium	Medium	Control positive:negative ratio (default 1:1). Apply class weights in training. Report results for multiple ratios.	Use oversampling (SMOTE for tabular, augmentation for sequences) if imbalance is severe.
DR-05	Data leakage through overlapping regions, shared regulatory domains, or chromosomal proximity.	Medium	Critical	High	Chromosome-holdout split. Verify no region overlap between splits. Check for data leakage using a "leakage detector" that tests whether split membership is predictable from features.	Leakage detected, re-design split. In worst case, use leave-one-chromosome-out evaluation.

3. Modeling Risks

ID	Risk	Likelihood	Impact	Severity	Mitigation	Fallback
MR-01	All models achieve < 0.70 AUROC: the task may be harder than expected, or features may be wrong.	Low-Medium	High	High	Check data pipeline for bugs first. Verify labels are correct. Try different feature sets. Ensure trivial baselines (random, GC-only) are reasonable.	If data and labels are correct and AUROC < 0.70, document as negative result. Pivot to multi-class (enhancer vs. promoter vs. background) which may have stronger signal.

MR-02	CNN does not improve over tree-based models: diminishing returns from deep learning at this scale.	Medium	Low	Low	Document as a valid finding. It means interpretable models are sufficient — this supports the project thesis.	Stay with tree-based models. This is a defensible research outcome.
MR-03	Pretrained embeddings do not improve classification: foundation model representations may not capture regulatory-relevant features at 1 kb scale.	Medium	Low	Low	Document as negative result. Compare embedding quality across different pretrained models.	Stay with Phase 2 results. Report that pretrained models didn't add value for this specific task and scale.
MR-04	Hyperparameter sensitivity: results change significantly with different hyperparameters, making findings unreliable.	Medium	Medium	Medium	Report best, worst, and median across a reasonable hyperparameter search. Use cross-validation for stability.	If results are unstable, increase dataset size or use ensemble methods.
MR-05	Overfitting (train AUROC ■ test AUROC).	Medium	Medium	Medium	Monitor train/val gap throughout training. Use regularization (L1/L2, dropout, early stopping).	Simplify model (fewer features, shallower architecture).

4. Interpretability Risks

ID	Risk	Likelihood	Impact	Severity	Mitigation	Fallback
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IR-01	Feature importance is dominated by GC content: the model relies primarily on overall nucleotide composition rather than specific motifs or patterns.	Medium	High	High	Include GC content as an explicit feature. If GC is top-important, run ablation with GC removed. Use GC-matched negatives.	If GC dominates even with matching, report this as a finding (composition is the primary signal). Add motif-scan features to capture finer-grained patterns.
IR-02	SHAP computation is too slow: for large feature sets (k=6, 4096 features) × many samples, SHAP may take hours.	Medium	Low	Low	Use TreeExplainer (fast for tree models). Subsample to 1K–5K samples. Use approximate SHAP methods.	Report feature importance from built-in model methods (Gini, permutation) instead of SHAP.
IR-03	CNN filters do not match known motifs: learned patterns may be biologically uninterpretable.	Medium	Medium	Medium	Visualize filters as sequence logos. Compare quantitatively (e.g., TOMTOM motif comparison). Some filters may capture non-motif structural features.	Report unmatched filters as "learned patterns" with appropriate caveats. Not every learned feature needs to match a known motif.
IR-04	Saliency maps are noisy or inconsistent: gradient-based attribution can be sensitive to input perturbations.	Medium	Medium	Medium	Use Integrated Gradients (more stable than vanilla gradients). Average over multiple reference baselines. Compare saliency patterns across samples.	Use SmoothGrad or ensemble saliency. Report limitations of gradient-based interpretability.

IR-05	Attention weights are misinterpreted: attention weights do not necessarily reflect feature importance, a known limitation in the interpretability literature.	Medium	Medium	Medium	If using attention, clearly state that attention weights are descriptive, not causal. Use attention only as supplementary to gradient-based methods.	Do not rely on attention as the primary interpretability mechanism.
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5. Compute Risks

ID	Risk	Likelihood	Impact	Severity	Mitigation	Fallback
CR-01	Kaggle reduces free-tier GPU quota: quota changes could limit weekly GPU hours.	Low-Medium	High	Medium	Monitor Kaggle announcements. Use CPU runtime for classical ML. Minimize GPU usage.	Migrate to Google Colab free tier. Consider Colab Pro (\$12/month).
CR-02	Kaggle session disconnects during long training runs, losing unsaved progress.	Medium	Medium	Medium	Save model checkpoints every 5 epochs. Save intermediate results to <code>/kaggle/working/</code> . Use try/except to save on error.	Re-run from last checkpoint. Design experiments to complete within 4 hours.
CR-03	Out-of-memory on Kaggle when loading pretrained models or large feature matrices.	Medium	Medium	Medium	Use float16 for pretrained model inference. Process in batches. Use sparse matrices for k-mer features.	Use smallest pretrained model variant. Pre-compute embeddings in multiple sessions.

CR-04	Library version conflicts on Kaggle: Kaggle's pre-installed packages may conflict with required versions.	Low	Low	Low	Pin versions in <code>requirements.txt</code> . Test on Kaggle early. Use <code>pip install --upgrade</code> for specific packages.	Adapt code to available versions.
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6. Timeline Risks

ID	Risk	Likelihood	Impact	Severity	Mitigation	Fallback
TR-01	Data pipeline takes longer than expected: parsing ENCODE formats, handling edge cases, debugging sequence extraction.	Medium	Medium	Medium	Start data pipeline work immediately (Phase 0). Use well-tested libraries (pybedtools, pysam). Test with small subsets first.	Allocate extra week. Use pre-processed datasets if available.
TR-02	Team member attrition: students leave the project mid-stream.	Medium	High	High	Comprehensive documentation (this pack). Modular code. Onboarding brief. No single points of failure.	Remaining team members can continue with documented codebase. Recruit replacements using onboarding doc.
TR-03	Scope creep: team members want to add features beyond current scope (multi-omics, variant analysis, web interface).	High	Medium	Medium	Scope lock in Charter and Glossary. Decision gates in Modeling Roadmap. All scope changes require documented justification.	Review scope lock document. Defer new ideas to Phase 4 or a separate project.

TR-04	Semester/academic calendar disruption: exam periods, breaks reduce available effort.	High	Medium	Medium	Build buffer into timeline. Design experiments that can be paused and resumed. Prioritize early milestones.	Accept slower progress during disruptions. Extend timeline by 1–2 months if needed.
TR-05	Delayed publication: results are not novel enough or reviewers raise issues.	Medium	Medium	Medium	Start writing early (background section during Phase 0). Target workshop papers first (lower bar). Plan for revisions.	Publish as preprint on bioRxiv. Present at internal seminars.

7. Risk Severity Summary

Severity	Count	Most critical risks
Critical	0 (none currently critical)	—
High	5	SR-01, SR-03, DR-05, MR-01, IR-01
Medium	14	Most modeling, data, and timeline risks
Low	5	Compute version conflicts, CNN not improving

Top 5 Risks to Monitor

1. **SR-03 / DR-05:** Data leakage or negative set bias → fundamentally undermines all results.
2. **SR-01 / IR-01:** Features capturing composition rather than regulatory patterns → undermines interpretability thesis.
3. **MR-01:** Models fail to achieve meaningful accuracy → blocks all downstream work.
4. **TR-02:** Team attrition → documentation and modularity are the primary mitigation.
5. **CR-01:** Kaggle quota reduction → have backup compute plan.

End of Document 09 — Risk Register

Next: [10 — Roadmap & Milestones](#)